

Comparative deep learning approaches for insulator condition monitoring in power transmission line

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ARTICLE INFO

Keywords:

Insulator condition monitoring
Deep learning approaches
Image Classification
Aerial Image Analysis

ABSTRACT

Accurate and timely condition monitoring of insulators can ensure the reliability and operational safety of power transmission systems. In this paper, a comparative study of deep learning-based approaches for identifying insulator condition using image data has been suggested. It categorizes the insulators into four classes such as broken, defect surface, polluted, and healthy. As the data set is not balanced, augmentation techniques are employed to balance the classes of the data-set. Features are extracted from the images using four deep learning models such as Autoencoder, InceptionV3, MobileNet, and Vision Transformer (ViT). Performance of the methods is evaluated using 5-fold cross-validation technique to enhance robustness and effectiveness. Among all the models used, the ViT approach outperforms others, achieving a validation accuracy of 99% and an F1-score of 0.99, owing to its superior spatial representation capabilities. Autoencoder yields an F1-score of 0.84 showing moderate performance. InceptionV3 achieves 60.78% accuracy and an F1-score of 0.66, likely due to over-fitting. The results indicate that ViT offers better performance and computational cost, making it a strong candidate for real-time identification of insulator condition in intelligent power grid monitoring systems.

1. Introduction

High-voltage power transmission lines condition and functionality is dependent on its essential components. Among the essential elements, insulator is a vital component that provide both mechanical support and electrical insulation between wires and transmission towers. Insulators are necessary for stability and protection. Insulators encounter environmental and mechanical stress, including intense heat, precipitation, pollution, snowfall, moisture, physical loading, and windborne particles. It will reduce insulation's efficiency and have a significant impact on the electricity grid's reliability. Transmission line must be inspected frequently, and any damaged insulator components must be replaced or repaired to ensure its safe and stable operation. Insulators are susceptible to faults, including breakage, self-explosion, and flashover that may be triggered by mechanical stress and climate changes. It can be avoided by promptly and accurately identifying these faults. Identification of insulator issues guarantee the power system's operational dependability and safety [1,2]. Traditional inspection methods can be inaccurate, time-consuming, labour-intensive, and prone to human error. Consequently, automated condition identification systems have become more vital in modern electrical power generation.

In the past few years, the utilization of artificial intelligence (AI), machine learning models (MLMs), and deep learning models (DLMs) such as convolutional neural network (CNN), Visual Geometry Group (VGG), autoencoder has proven highly effective in automating the assessment of insulator condition. Researchers in [3] provided an inclusive analysis of DLM for insulator fault detection, highlighting the improvements and challenges in this field of research. In [4], a CNN with dense connectivity and well-organized attention mechanisms was applied. In [5], the effectiveness of CNN based on image features for reliable fault identification is demonstrated but there exists a class imbalance problem. In [6], aerial imagery was leveraged in conjunction with VGG16 deep learning models to detect insulator breakdowns, providing a robust foundation for large-scale deployment. In [7], A deep convolutional auto-encoder was developed for fault recognition in scenarios with limited negative samples, demonstrating promising results. An anomaly identification framework using auto-encoder was presented in [8] to improve the precision of automated insulator inspections.

Numerous works have investigated sophisticated deep learning techniques for identifying insulator faults, with a specific emphasis on YOLO-based models. In [9], YoLoV3 has been applied for insulator fault detection in diverse aerial images. In [10], cross stage partial dense

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Table 1
Summary of existing methods.

Ref.	Techniques Used	Class Imbalance exist	(Accuracy/ mAP@0.5) > 98%	Complexity of network more	Key Strengths	Key Limitations / Gaps
[5]	CNN	Yes	No	Yes	Demonstrate DL feasibility	Comparative depth limited; lacks unified evaluation protocol
[9]	YOLOv3	Yes	No	Yes	Enables reproducibility	Dataset bias; limited condition diversity
[11]	State space model-guided YOLO	Yes	No	Yes	Superior multi-scale feature fusion	Computationally expensive
[12]	YOLOv5 network	Yes	No	Yes	Real-time capability	Emphasis on detection speed over condition severity
[13]	Hybrid YOLO using ResNet-18 classifier	Yes	Yes	Yes	Improved classification robustness	Limited scalability across datasets
[14]	YOLOu-Quasi-ProtoPNet	Yes	No	Yes	Improved accuracy	High architectural tuning
[16]	DCN-ELAN backbone network	Yes	No	Yes	Lightweight deployment	Minimal analytical comparison between YOLO variants
[17]	, YOLOv8-IDX	Yes	No	Yes	Advanced YOLO model	Less emphasis on condition severity
[18]	YOLOv8	Yes	No	Yes	Improved feature representation	Model complexity high; limited explainability; horizon or defect-severity sensitivity not analyzed
[19]	Multi-scale feature fusion transformer model	Yes	No	No	Future Fusion	Lacks benchmarking against classic CNN/YOLO under identical conditions
[22]	An ensemble of deep learning model	Yes	Yes	Yes	Improved detection accuracy	Increased inference time; interpretability concerns

YOLOv3 has been applied for improved insulator fault detection by utilizing a DLM designed explicitly for various aerial images of high-voltage lines. In [11], a state-space model guided feature perception YOLO network is proposed for high-resolution aerial imagery. In [12], an ML-YOLOv5 algorithm is presented that improves detection accuracy by refining feature extraction mechanisms. Ref [13] presented a hybrid-YOLO model that combined ResNet-8 with YOLOv5 for drone-based monitoring, exhibiting enhanced classification effectiveness in actual aerial images. In [14], this field was further advanced by utilizing an optimized hybrid YOLOu-Quasi-ProtoPNet framework, which achieved higher precision in insulator categorization. The lightweight YOLOv5 variant [15] and the improved YOLOv7 framework [16] have addressed the computational efficiency and complexity of drone-based image detection of insulators. More recently, YOLOv8-IDX [17] and modified YOLOv8 networks [18] have further enhanced precision, robustness, and the handling of data imbalance.

The current studies have focused on enhancements that integrate attention and transformer designs for the identification of insulator faults. In contrast, [19] presents a multi-scale feature combination transformer with hybrid attention to precisely identify insulator faults. Similarly, in [20], a Hierarchical Rotation Gaussian Attention Network (HRGA-Net) is introduced for accurate drone-based insulator detection. Some of the studies also combined different deep learning models to increase the performance in detection. In [21], a hybrid DLM called CNN-LSTM for bright aerial monitoring of transmission line insulators was proposed, improving detection efficiency under diverse conditions. In [22], optimized ensemble of deep learning models (OEDL) based on weighted boxes fusion has been proposed for improved fault detection.

Summary of some of the existing works on insulator condition monitoring has been noted in Table 1 focusing on class imbalance problem, complexity of the network and performance of the method. Although extensive research as tabulated in Table 1 has explored deep learning-based insulator defect detection using CNNs, YOLO variants, attention mechanisms, and transformer-based architectures, the existing literature largely remains model-centric and application-specific. Most studies evaluate single or closely related architectures under customized datasets, making cross-model performance comparison difficult. Further, current works primarily focus on binary defect detection, with limited attention to condition severity differentiation, robustness across varying inspection scenarios, and systematic benchmarking under identical experimental settings. Attention-enhanced models demonstrate promising accuracy but incur high computational complexity.

YOLO-based approaches prioritize real-time detection but it has the expense of analytical interpretability.

All the above described methods have advancements in insulator fault identification but several research gaps that still exist are identified as below:

- I. Most of the methods face challenges because of the image quality, occlusion, and complex backgrounds.
- II. The dataset available has class imbalance between normal and faulty samples that will lead to incorrect accuracy.
- III. Existing methods has less accuracy in predicting the class of insulator faults.
- IV. Achieving balance between recognition accuracy and computational efficiency remains a challenge for real-time deployment.
- V. A comparative framework that evaluates diverse deep learning paradigms, under a uniform dataset and consistent evaluation protocol has not been suggested.
- VI. Limited emphasis given on condition monitoring, severity assessment, and decision support required by power utilities for maintenance planning and asset management.
- VII. Inaccurate identification of insulator condition may lead to more power outage time in the power system.

This work proposes a comparative analysis of deep learning-based techniques to address these challenges, focusing on improved feature extraction, robust identification under different conditions, and computational efficiency to support the reliable monitoring of insulators in modern power transmission systems. The proposed work utilizes insulator images as input and employs advanced deep learning models to extract discriminative features and classify images into appropriate categories. The major contributions of this research are summarized below:

- A comparative deep learning framework is developed for insulator fault detection using images.
- Data augmentation is applied to mitigate class imbalance problem.
- A comparative analysis of several deep learning models like Autoencoder, InceptionV3, MobileNet, and ViT is carried out for classifying insulator conditions.
- The ViT model achieves the highest performance with 99% validation accuracy and an F1-score of 0.99, demonstrating strong spatial feature representation and computational efficiency.

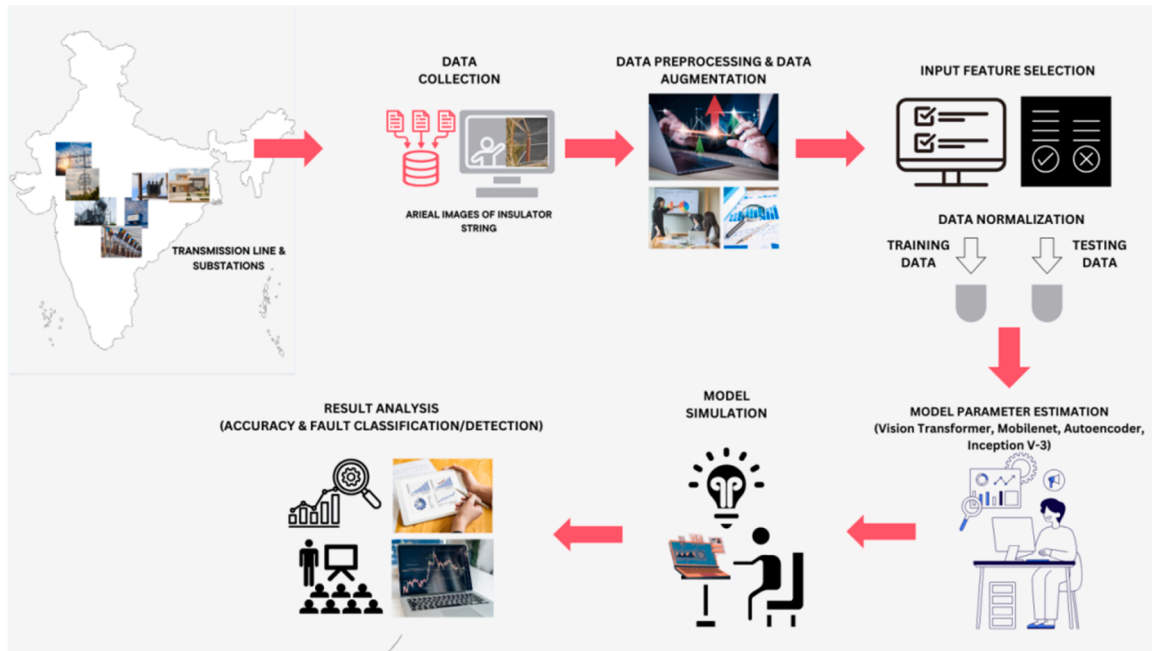


Fig. 1. Framework of the proposed method.

- The ViT-based approach offers reliability and reduced complexity, making it suitable for real-time insulator condition monitoring.
- To the best of the knowledge, ViT has not previously been applied to insulator fault condition monitoring.
- The proposed image-based insulator fault identification approach, outperforms several state-of-the-art techniques.
- As the method accurately identify the condition of insulator, any problem in insulator can be identified immediately and the repair work can be done. As a result, the power outage time will be reduced.

The Section 1 of the work introduces importance of insulator condition monitoring in transmission systems. Section 2 describes the problems associated with transmission line insulator. Section 3 describes insulator datasets, pre-processing and augmentation techniques to address class imbalance. Section 4 outlines the proposed methodology, describes the deep learning models like Autoencoder, InceptionV3, MobileNet, and ViT. Section 5 presents the experimental findings, model assessments, discussions and performance comparisons. Section 6 concludes the study and suggest potential avenues for further research.

2. Problems associated with transmission line insulator

Power transmission lines insulators providing high electrical resistivity, mechanical strength and support. Insulators are made from materials like porcelain, glass, or polymer, they are crucial for the safe and efficient operation of overhead power lines. Insulators are of several types, including pin, suspension, strain, and shackle insulators, each suited for different voltages and applications. The porcelain insulators of suspension and strain types are discussed here in this work. Different defects may happen in transmission line insulators as it is always exposed to environment. In this work, only porcelain insulators images are considered for fault diagnosis. The insulators can be categorized into different types based on the conditions associated. Fig.1 shows the images of the possible types of insulator defects such as broken, defect surface, polluted and as well as healthy insulator; and discussed in detail as follows:

- Broken insulators is an insulator which has lost its dielectric property due to external over-voltage or mechanical fracture.

- Defect surface insulators are the insulators where an electrical discharge has occurred along the surface, a phenomenon known as a surface flash-over. This flash-over happens when the normal insulating properties of the surface are overcome, often due to a combination of pollution (like salt, dust, or chemical residues), moisture, and the formation of dry bands on a polluted surface.
- Polluted insulators are those where conductive layers of dirt, salt, and moisture accumulate on the surface of electrical insulators.
- Healthy insulators are free from defects such as cracks, breakage, and surface pollution.

3. Proposed method

The proposed method focuses on developing an automated condition monitoring and classification system for transmission line insulators using deep learning techniques. The proposed method comprises key stages, including database preparation and pre-processing, data augmentation, model architecture and training, evaluation, and performance analysis. Fig.1 shows the framework of the proposed method. Furthermore, a detailed process flowchart has been prepared and illustrated in Fig.2 which gives the complete understanding of the steps followed in developing the proposed scheme. The individual steps are discussed in detail in subsequent sub sections.

3.1. Inputs used

The proposed method has been validated using various types of images as input taken from the IEEE data port [23]. The images of insulators are categorized into four classes: broken insulators, defect surface insulators, polluted insulators, and healthy insulators. Fig.3 illustrates the original images for all four classes.

3.2. Pre-processing of insulator images

Since deep learning techniques require large amounts of data to generalize effectively, the data-set is preprocessed to ensure consistency and quality. The images are re-sized to a uniform shape of 152×152 pixels for Autoencoder, Inception and MobileNet model and 224×224 pixels for ViT to maintain computational efficiency. The images of

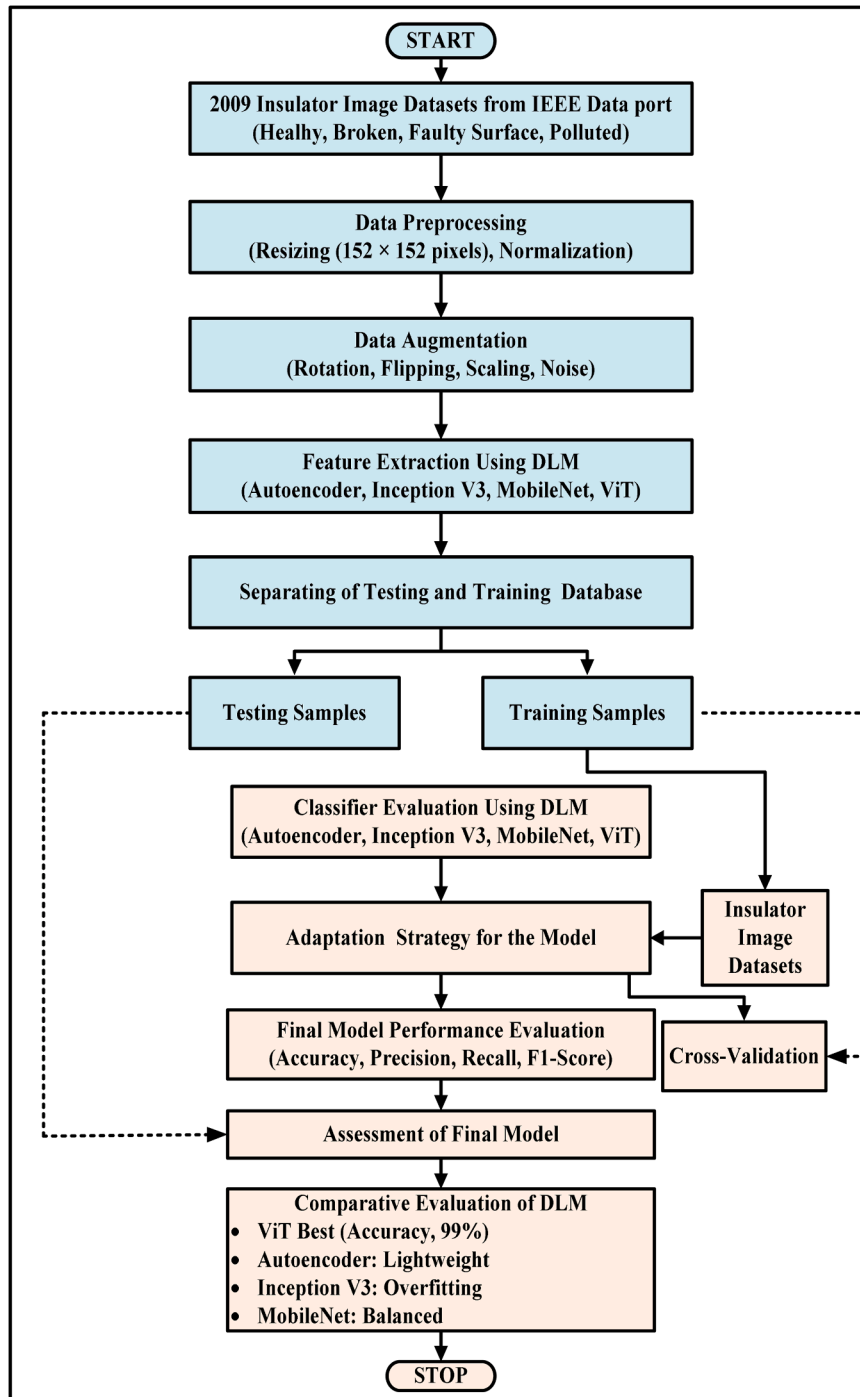


Fig. 2. Flowchart of the proposed method.

different classes are not equal in number, which may cause a class imbalance problem. To avoid data leakage, k-fold stratified sampling with cross validation (4-fold) is used that split the data first into distinct training, validation, and test sets to maintain the original class distribution across all subsets. Then augmentation method has been applied for class balancing in training, testing and validation separately. In the next section, the class balancing using image augmentation have been discussed.

3.3. Image augmentation for class balancing

To address the problem of class imbalance and enhance the model's

generalization capacity, data augmentation is employed. A structured data augmentation method was applied using the Keras ImageDataGenerator module. A goal of recording 1000 images per class has been proposed to achieve an even distribution. To improve the model's ability to learn representative characteristics from each class, the goal has been established to balance the data-set. The ImageDataGenerator module is equipped with augmentation parameters that enable the synthetic production of additional image samples by applying diverse modifications to the current images. The subsequent transformations were implemented:

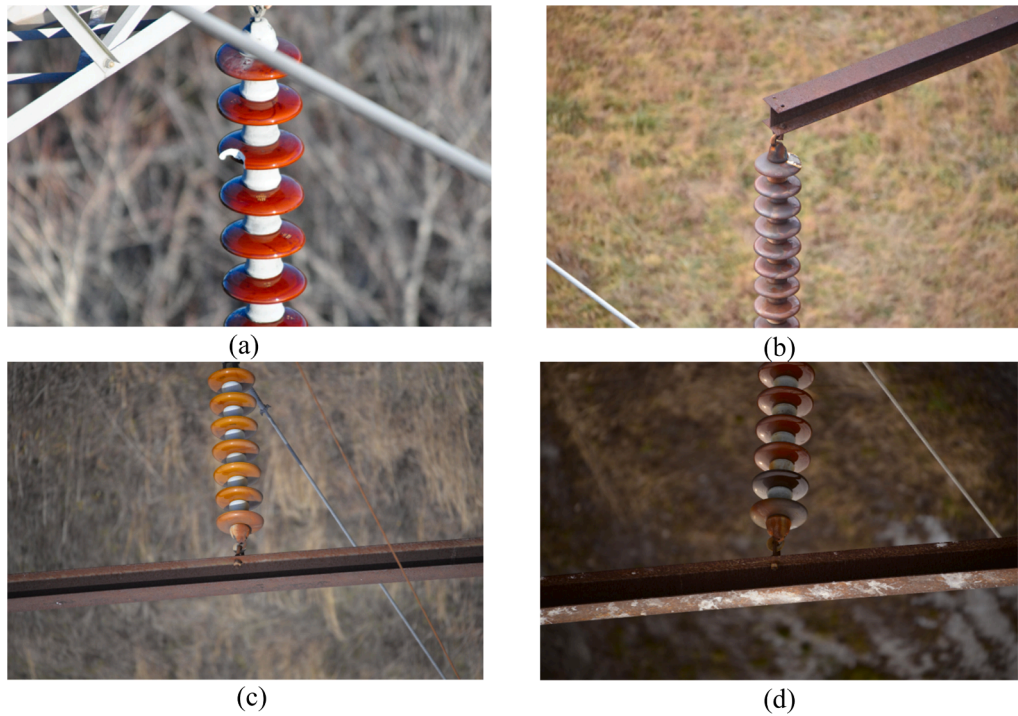


Fig. 3. Insulator images (a) Broken (b) Defect surface (c) Healthy (d) Polluted.

Table 2
Details of the image datasets before and after augmentation.

Image Classes	Number of images	
	Before Augmentation	After Augmentation
Broken	983	1881
Defect Surface	875	1760
Healthy	109	1035
Polluted	42	998
Total	2009	5674

- I. Rotation range: Images underwent random rotation within an extensive selection of ± 20 degrees to accommodate angular variations.
- II. Width and height shifts: Spatial translation by applying horizontal and vertical shifts of up to 20% of the image dimensions.
- III. Shear transformation: A shear transformation with an intensity of 0.2 has been used to replicate minor geometric distortions in the photos.
- IV. Zoom transformation: Random zooming within a 20% range has been effective in generating varied magnified versions of the images.
- V. Horizontal flipping: Images have been randomly mirrored along the horizontal axis to generate flipped versions, enhancing variability.
- VI. Fill mode: The 'nearest' fill mode is used to replace vacant areas formed during geometric transformations with the values of the closest neighbouring pixels.

The augmentation process is implemented across all four class

through an iterative looping approach. After augmentation, each class record had at least 1000 images making it nearly a total of 5674 images. These techniques present random variations to current images while conserving their original class labels. The number of images before and after augmentation is shown in Table 2. These transformations improve the model's generalization capability and improve performance on unseen data.

3.4. Techniques used

Various deep learning models used for insulator fault detection have been discussed here. The models used include Autoencoders, InceptionV3, MobileNet, and ViT for feature extraction and classification tasks. All the models used are explained individually hereunder subsequent subsections.

3.4.1. Autoencoder

An autoencoder learns essential patterns by encoding the input into a reduced dimensional representation and then reconstruct to match the original input. The architecture consists of two key components: an encoder, which transforms input data into a compressed latent representation, and a decoder, which reconstructs the original data from this compact form. In this research, these models (DLMs) are developed for classifying insulator defects, and the constructed features are used to distinguish between different insulator conditions accurately. Convolutional encoder composed of multiple convolution layers followed by max pooling layers. These layers extract and compress the spatial features into a bottleneck (latent space). The convolutional decoder consists of up-sampling and convolution layers that expand the latent space representation back to the original image dimensions.

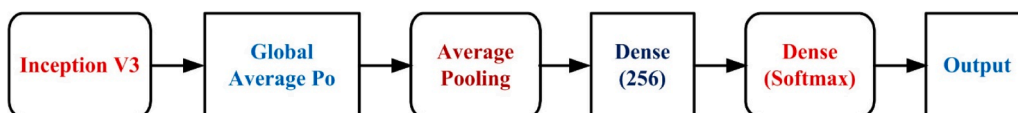


Fig. 4. Architecture of inception V3.

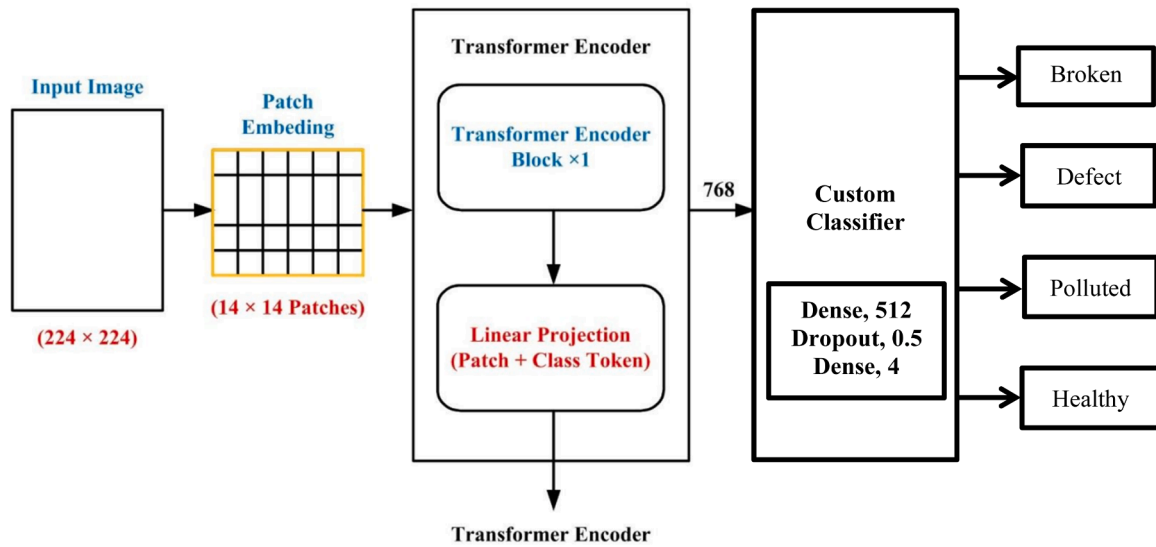


Fig. 5. Vision Transformer (ViT) architecture.

3.4.2. Inception V3

Inception V3 is an advanced deep CNN architecture that builds upon the original inception (GoogLeNet) model. By leveraging parallel convolution blocks of varying kernel sizes within the same layer, the inception family aims to capture both fine and coarse visual patterns effectively. The key architectural components of the Inception v3 network are illustrated in Fig. 4. Factorized convolutions are decomposed into two or more smaller operations (e.g., 1×5 followed by 5×1). It drastically reduces the number of parameters while preserving the receptive field. Auxiliary classifiers include auxiliary outputs during training to improve gradient flow and prevent vanishing gradients in deeper networks. Asymmetric convolution filters are used to make computation more efficient without losing the ability to extract spatial features. Inception V3 uses stride-based convolution for spatial dimension reduction rather than pooling layers, which helps retain information while reducing the feature map size. At the end of the network, global average pooling replaces traditional fully connected layers, thereby decreasing overfitting and improving generalization.

3.4.3. MobileNet

MobileNet is a family of lightweight deep CNNs explicitly designed for efficient performance on mobile and embedded devices. The core idea behind MobileNet is to decompose the standard convolution operation into two separate layers: a depth-wise convolution, which

applies a single filter per input channel, and a pointwise convolution (also known as a 1×1 convolution), which combines the outputs from the depth-wise layer. MobileNet models are highly scalable used for various tasks, including image classification, object detection, and segmentation. Their lightweight design makes MobileNet a powerful choice for real-time applications.

3.4.4. Vision transformer

ViT is a deep learning architecture that applies the transformer model. ViT model has various advantages over traditional CNN model. ViTs capture global context via self-attention, offering better scalability on large datasets, superior performance in complex tasks such as imaging, and strong transfer learning, while CNNs may not. Self-attention allows ViTs to weigh the importance of all image patches at once, capturing global relationships from the start, unlike CNNs that build context layer-by-layer making it more complex. ViTs can work better with massive datasets while CNNs utilize more computational power as data increases. ViT excel at adapting pre-trained knowledge to new tasks with less labeled data in comparison to CNN. Patching reduces initial memory usage compared to large CNN activation maps, aiding batch size. In view of the advantages of ViT over other techniques, it is utilized for insulator condition monitoring task in this paper.

Vanilla ViT that is a variant that is pre-trained on ImageNet. The pre-training weights are taken from there. Key components of the ViT

Table 3
Training performance of various methods.

Method	Network training hyper-parameters	Epoch	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
Inception V3	Learning rate - 0.001, Optimizer-Adam, Batch size - 10, Activation function - ReLU and SoftMax, Early stopping	10	91.70%	0.199	59.17%	2.318
		20	93.19%	0.177	60.53%	2.048
		30	95.20%	0.129	60.50%	2.018
		40	95.68%	0.119	60.78%	2.010
Auto-encoder	Learning rate - 0.001, Optimizer-Adam, Batch size - 10, Activation function - ReLU and Sigmoid/Linear, Early stopping	10	85.51%	0.594	84.98%	0.586
		20	86.17%	0.592	85.57%	0.584
		30	86.27%	0.590	85.60%	0.583
		40	86.90%	0.572	85.68%	0.560
MobileNet	Learning rate - 0.001, Optimizer-Adam, Batch size - 10, Activation function - ReLU and SoftMax, Early stopping	10	78.49%	0.717	48.59%	1.336
		20	83.48%	0.492	52.80%	1.509
		30	86.52%	0.475	51.76%	1.638
		40	88.20%	0.355	55.20%	1.659
Vision Transformer (ViT)	Learning rate - 0.001, Optimizer-Adam, Batch size - 10, Activation function - ReLU and SoftMax, Early stopping	10	97.43%	0.070	97.12%	0.076
		20	99.22%	0.020	97.89%	0.068
		30	99.00%	0.032	98.68%	0.063
		40	99.04%	0.031	98.67%	0.063

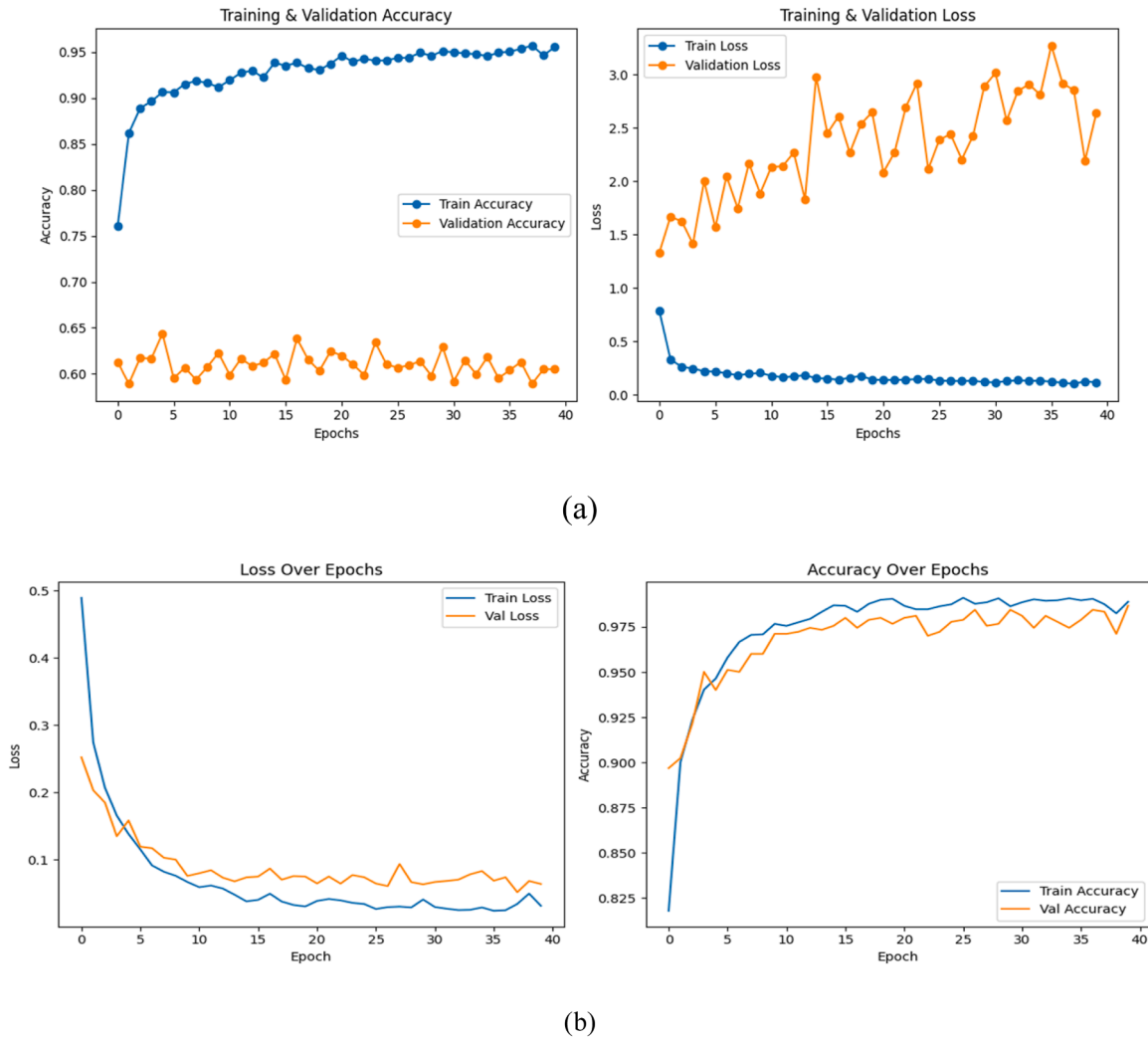


Fig. 6. Performance of different methods (a) Inception V3 (b) Vision Transformer (ViT).

architecture are shown in Fig. 5 and can be described as follows. Patch embedding involves splitting the input image into non-overlapping patches. Then each patch is converted into a one-dimensional vector. Positional encoding is added to each patch to retain spatial relationships. Multi-head self-attention blocks allow the model to focus on relevant areas within the image. The classification token serves as a representative summary of the entire image, used for classification purposes. MLP head (classifier) is a fully connected layer stack used for final prediction based on the transformer output.

3.5. Development of insulator condition identification framework

The features from the insulator images have been extracted using various deep learning techniques. After the features are extracted from the images, it is given to the classifier to identify the insulator condition. The details of the insulator condition identification techniques using different deep learning models are discussed below.

3.5.1. Auto-encoder

The performance of the autoencoder along with the network specifications to identify the insulator conditions in terms of accuracy and loss across different epochs is shown in Table 3. Since auto-encoders learn a compact representation of normal (healthy) insulators, they can effectively identify other types based on deviation from standard patterns. Key training parameters used are the mean square error (MSE)

loss function, the Adam optimizer, 40 epochs, a batch size of 10, and activation functions of ReLU in the hidden layers and sigmoid or linear in the final decoder layer.

3.5.2. InceptionV3

The pre-trained InceptionV3 model is fine-tuned on the insulator condition images, allowing it to learn domain-specific features while retaining its powerful generalization capabilities. The performance of the inceptionV3 model, in terms of accuracy and loss, across different epochs is illustrated in Fig. 6 and network parameters are summarized in Table 3. Inception V3 is utilized as a feature extractor due to its capability to handle complex and high-resolution image features. The base Inception V3 model was loaded without its top layers, allowing for customization for domain-specific tasks. The output from the convolutional base was passed through a GlobalAveragePooling2D layer, followed by a dropout layer (for regularization), and then a dense SoftMax classification layer tailored for four classes.

3.5.3. MobileNet

In this study, MobileNet was implemented to identify the condition of insulators into four categories of high-voltage insulators. The images are preprocessed to a fixed resolution of 224×224 pixels, converted to a 3-channel format to match MobileNet input specifications. The model with network parameter as depicted in Table 3 was trained over 10, 20, 30, and 40 epochs, and the performance was monitored through

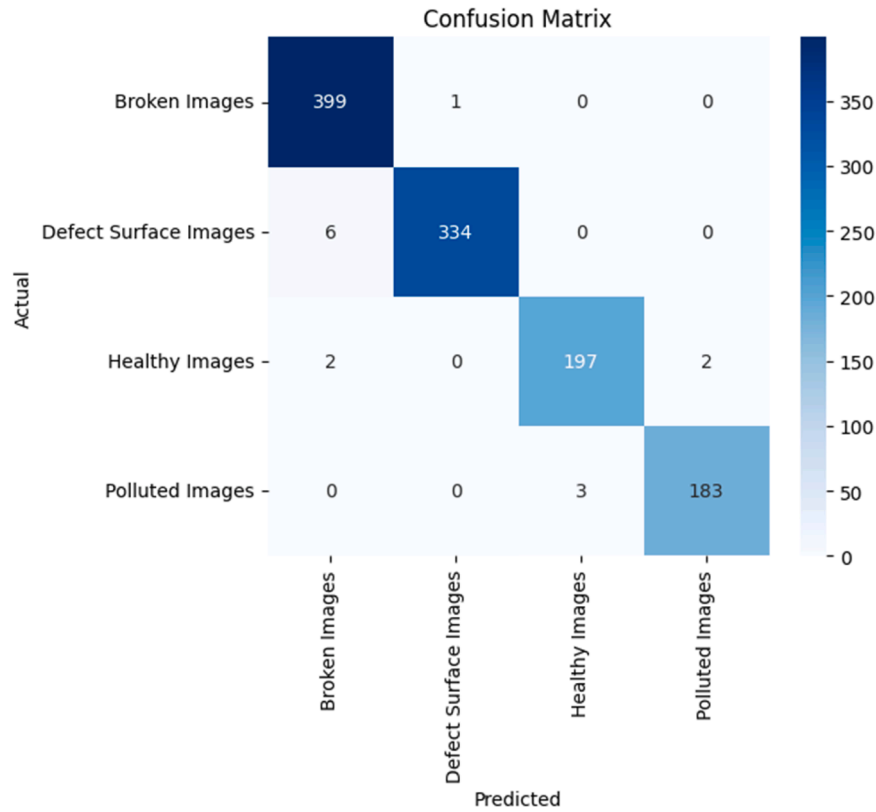


Fig. 7. Confusion matrix of ViT.

accuracy, loss, and F1-score metrics. Despite the computational efficiency of MobileNet, the model exhibited relatively lower performance in this domain-specific task, with a validation accuracy of 55.20% peaking after 40 epochs. The performance of the MobileNet model is shown in Table 3.

3.5.4. Vision transformer (ViT)

In this work, the ViT architecture was adapted and implemented for the automated identification of insulator condition. Each image was resized to 224×224 to match the input requirements of ViT. Since ViT models expect 3-channel input, each grayscale image was expanded into three identical channels. The images were divided into fixed-size patches using a pretrained ViT backbone (vit-base-patch16-224). A ViT image processor handled pixel standardization and positional embedding. Feature extraction was performed using the output of the class token from the final transformer layer, yielding a 768-dimensional feature vector per image. Extracted features were passed through a sequential MLP classifier comprising two fully connected layers (Dense (512), Dense (256)) with dropout layers to prevent over-fitting. The final layer used a soft-max activation for 4-class classification. The model was trained over 40 epochs without early stopping to ensure complete learning. Optimization was performed using the Adam optimizer with learning rate scheduling. The performance of the ViT model in terms of accuracy and loss across different epochs is shown in Fig.6 and Table 3.

4. Experimental results

The proposed approach uses deep learning techniques for insulator condition identification. It provided an automated solution for improving power system reliability. The performance metrics includes accuracy, loss, precision, recall, and F1-score. The experiments are conducted on the Windows operating system. The CPU model is 12th Gen Intel(R) Core (TM) i7-1270, featuring a clock speed of 2.10 GHz and 8GB of RAM. The model building, training, and assessment have been

Table 4
Analysis of accuracy.

Classes	Auto-encoder	Inception V3	MobileNet	ViT
Broken	82.50%	68.75%	33.75%	99.75%
Defect	84.72%	69.44%	33.33%	98.23%
Healthy	85.57%	61.19%	12.43%	98.00%
Polluted	83.33%	64.51%	26.8%	98.38%

conducted using Python-based deep learning and transformer libraries, such as Hugging Face's transformers and TensorFlow. All implementation work has been done using Python on the Google Colab platform. The trained models are evaluated on the test input images, and their performance is analyzed using multiple metrics such as accuracy, precision, recall, and F1-Score that provide insights into model robustness for each class. A comparative analysis is conducted to identify the most effective model for classifying insulator conditions.

4.1. Analysis of accuracy of condition monitoring of insulator

The accuracy of the proposed method for condition monitoring of insulator has been discussed. Accuracy measures the overall correctness of the model's predictions. To calculate the accuracy, first the confusion matrix is obtained. The confusion matrix of the ViT method has been shown in Fig.7. The accuracy of all the methods class-wise has been shown in Table 4. It can be observed that the ViT model has highest accuracy among all methods. One image from broken class is classified as defect class incorrectly. Six images from defect class are classified as broken class incorrectly. Two images from healthy class are classified as broken class incorrectly. Two images from healthy class are classified as polluted class incorrectly. Three images from polluted class are classified as healthy class incorrectly.

Table 5

Analysis of precision with various network.

Classes	Auto-encoder	Inception V3	MobileNet	ViT
Broken	0.82	0.68	0.34	0.98
Defect	0.85	0.72	0.30	1.00
Healthy	0.86	0.63	0.16	0.98
Polluted	0.85	0.66	0.26	0.99

Table 6

Analysis of sensitivity in predicting insulator fault or recall.

Classes	Auto-encoder	Inception V3	MobileNet	ViT
Broken	0.83	0.71	0.33	1.00
Defect	0.84	0.68	0.37	0.98
Healthy	0.87	0.60	0.10	0.98
Polluted	0.83	0.62	0.29	0.98

Table 7

Analysis of F-score for various models used.

Classes	Auto-encoder	Inception V3	MobileNet	ViT
Broken	0.82	0.69	0.34	0.99
Defect	0.84	0.70	0.33	0.99
Healthy	0.86	0.61	0.12	0.98
Polluted	0.84	0.64	0.27	0.99

4.2. Analysis of positive prediction of condition monitoring of insulator

The positive prediction of insulator condition monitoring or precision of all the types of insulator defects has been noted in Table 5. The auto-encoder model was employed to learn compressed feature representations of insulator images with a precision of 0.82 to 0.86 considering all classes. The precision of MobileNet and Inception V3 are lower than other two network. ViT model has highest precision ranging between 0.98 and 1.00. Hence, the ViT network has better prediction in positive insulator condition monitoring.

4.3. Analysis of sensitivity of condition monitoring of insulator

The sensitivity of the insulator condition monitoring with different deep neural network has been noted in Table 6. The sensitivity of auto-encoder is between 0.83 to 0.87 considering all classes. Inception V3, a deep CNN known for its architectural efficiency and ability to extract multi-scale features, was utilized in this study to classify insulator images into four distinct fault categories with sensitivity between 0.60 to 0.71. MobileNet model has lowest sensitivity among all the implemented model. ViT model has highest recall ranging between 0.98 and 1.00.

4.4. Analysis of harmonic mean or F-score of condition monitoring of insulator

The F-score of the insulator condition monitoring method has been noted in Table 7. The auto-encoder model achieved its best performance in identifying healthy with an F1-score of 0.86, followed closely by defect and polluted. Using inception v3, defect surface images class achieved the highest F1-score of 0.70, followed by broken at 0.69. The model demonstrated the lowest performance in classifying healthy, with an F1-score of 0.61, likely due to the subtle visual differences between healthy and mildly polluted or defective, which can be challenging to distinguish even for advanced models.

MobileNet method F1-score across all classes showed limited effectiveness in distinguishing between subtle insulator faults. This outcome suggests that while MobileNet is advantageous in terms of speed and memory footprint, it may lack the depth and representational power

Table 8

Performance of all techniques.

Models	Precision	Recall	F1-score	Accuracy
Inception V3	0.67	0.65	0.66	61.00%
Auto-encoder	0.85	0.84	0.84	86.00%
MobileNet	0.27	0.27	0.27	29.00%
ViT (Proposed Technique)	0.99	0.99	0.99	99.00%

required for complex fault classification tasks involving high-resolution transmission line images. It also indicates need for model fine-tuning, data augmentation, or domain adaptation. ViT captured subtle variations in complex image conditions with F-score between 0.98 to 0.99. ViT model performed well even with a moderately sized data-set.

4.5. Discussion

The proposed method evaluates and compares the effectiveness of deep learning models for identification of insulator conditions. The overall performance of each technique has been analyzed and summarized in Table 8. The Autoencoder-based method delivered moderate performance, achieving precision of 0.85, recall of 0.84, F-score of 0.84 and an overall accuracy of 86%. The InceptionV3 network exhibited high training accuracy of 96% but significantly lower testing accuracy of 61%. The precision of 0.67, recall of 0.65 and F-score of 0.66. This indicates over-fitting, suggesting that the model learned patterns present in the training set but unable to identify unseen data. Similar result of low performance also happened with Mobilenet model. The ViT model showed precision, recall, and F1-score values consistently at 0.99 and overall accuracy of 99%. ViT model performs consistently across all categories, making it highly suitable for deployment in power system monitoring applications. The better performance of ViT network comparison to others is due to the following reasons:

- ViT uses self-attention mechanisms allows the model to differentiate between fault types such as broken, polluted, or defective surfaces.
- ViT models excel in insulator fault detection over CNNs because it captures global context, understanding relationships across the entire image, unlike CNNs' local focus, making them better at spotting small, varied defects amidst complex backgrounds and noise.
- The observed superior performance of ViT is justified through its ability to handle scale variation, background complexity, and structural continuity, strengthening the interpretability.
- By treating images as sequences of patches and weighing their importance, ViTs offer superior feature representation, leading to higher accuracy.

The empirical findings of the insulator condition identification method can be outlined as follows:

- An automated image-based detection of insulator defects is proposed for maintenance and reliability of power system network
- A comparative analysis of several deep learning models is carried out for classifying insulator images under varying conditions.
- The ViT model demonstrated strong classification performance achieving precision, recall, and F1-scores of 0.99, and an average accuracy of 99%.
- ViT's balanced class-wise metrics indicate its ability to extract relevant spatial features from image data, making it suitable for real-time condition monitoring of insulators.

4.6. Comparison of the proposed method with existing techniques

In order to evaluate the performance of the ViT-based insulator condition monitoring framework, a detailed comparison was carried out against several existing deep learning-based approaches. The evaluation

Table 9

Summarizes different classifier models and their corresponding performance.

Author	Dataset	Dataset Name	Techniques	Performance	Remarks
[7]	Different	Insulator Images	Autoencoder	Accuracy rate of 88%	Effective on limited samples using an Auto-encoder
[8]	Different	Insulator Dataset	Enhanced CNN	Accuracy rate of 91.3%	Improved inspection automation
[10]	Different	Chinese Power Line Insulator Dataset	Deep Model	Accuracy rate of 92%	Transmission line fault detection
[6]	Different	Aerial Insulator Images	Deep Learning-Based Inspection	Accuracy: 96.9%; AP50: 95.5%	Robust identification of insulator breakdowns using aerial imagery.
[4]	Different	Chinese Power Line Insulator Dataset	CNN with Dense Connectivity & Attention	Accuracy rate of 92.5%	Better learning with connectivity
[19]	Same	IDID	Multi-scale feature fusion transformer (MSFFT) model	mAP50:95 of 80.9%	Exceptional precision in detecting the most minor defects in insulators
[9]	Same	IDID	YOLO	mAP@.5 of 97.27%	Detect the multi-scale insulators by capturing different-scale and multi-view details
[11]	Same	IDID	State space model-guided deep feature perception network	mAP @0.5 of 81.8%, with only a marginal decrease of 0.5%	Address ultra-small defects or classes with severely scarce samples
[5]	Same	IDID	Deep Neural Network	Accuracy rate of 96.24%	Robust classification of image-based insulator faults
[22]	Same	IDID	An ensemble of deep learning model	mAP@[0.5] of 0.9841	Reduces the chance of error
[12]	Same	IDID	YOLOv5 network	mAP0.5 of the model increased by 1.4%.	Reductions in parameter count and FLOPs, down to 3.73 M and 9.
[13]	Same	IDID	Hybrid YOLO using ResNet-18 classifier	F1_score is 0.96216	Identification of conditions of transmission insulators
[14]	Same	IDID	YOLOu-Quasi-ProtoPNet	F1-score higher than 0.97	Electrical grid inspections
[15]	Different	Not Mentioned	Improved YOLOv5s with CBAM Attention	mAP: 95.35%; Small Defect Accuracy: 93.32%	Enhanced detection of small-scale defects with attention mechanisms.
[16]	Same	IDID	DCN-ELAN backbone network	Accuracy 81.6%	Simplifies the model's complexity and reduces computational requirements
[17]	Same	IDID	, YOLOv8-IDX	F1 scores of 93.2%	Automating power line inspection
[18]	Same	IDID	YOLOv8	mAP0.5 by 7.5% over the baseline	Strengthen the link between the localization and classification branches
Proposed Model	Same	IDID	ViT	Best Accuracy: 99.0%, F1-score: 0.99	ViT showed the best balance of accuracy, complexity, and training time

parameters are classification accuracy, model complexity, ability to generalize across diverse operational conditions, and computational efficiency. Traditional CNNs and their enhanced variants exhibit limitations in capturing global dependencies and long-range contextual relationships. The transformer-based architecture offers superior representation learning through self-attention mechanism. The comparison summary is presented in Table 9 that outlines the data-set used (e.g. Insulator Defect Image Dataset (IDID), methodological approach, and evaluation metrics. It also includes the remarks highlighting the unique contributions and limitations of each technique. Earlier models such as autoencoders, CNNs, and hybrid YOLO frameworks achieved considerable success with accuracies ranging from 88% to about 97%.

5. Conclusion

The proposed work uses deep learning-based techniques for insulator condition monitoring. A comparative analysis of several deep learning models such as Autoencoder, InceptionV3, MobileNet and ViT is carried out for classifying insulator images under varying conditions. Out of all the deep learning techniques used, ViT perform better than Autoencoder, InceptionV3, MobileNet. The ViT model has validation accuracy of 99% and F1-score of 0.99. ViT model has moderate model complexity and faster training time than others. Autoencoder offered moderate performance but lacked power to classify defect variations. InceptionV3 model exhibited poor performance due to domain mismatch and overfitting. The study focuses on the critical role of technique selection in insulator condition monitoring tasks. It suggests that ViT is the most practical and reliable technique for real-world deployment for insulator condition monitoring. Future work will focus on integrating multimodal data sources, real-time deployment in drone-based monitoring systems, and fine-tuning technique to enhance performance in smart grid

infrastructure.

CRedit authorship contribution statement

Anamika Yadav: Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Abhinav Kumar Nirala:** Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Aleena Swetapadma:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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